

TOA ASSIGNMENT 3

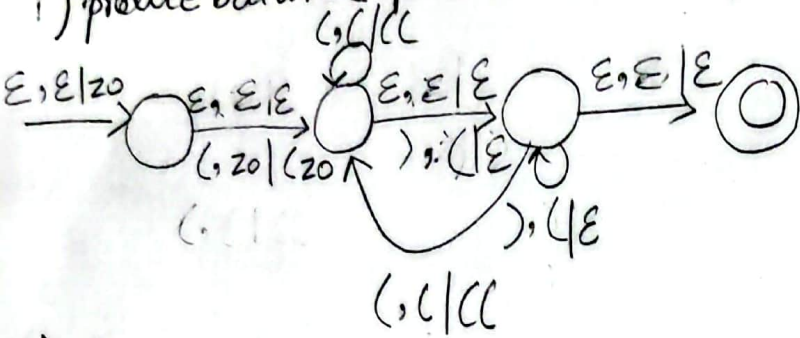
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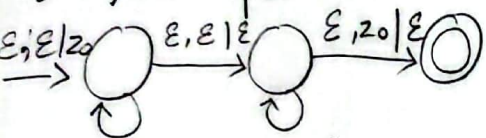
3F.

Q5 Design PDA's (CFG done in Assignment 1)

i) produce balanced parenthesis for $\{ (,) \}$.

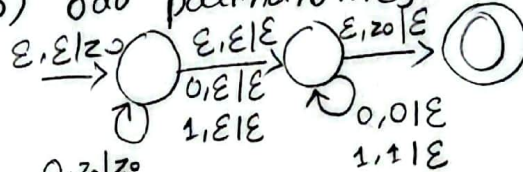


ii) a) even palindromes

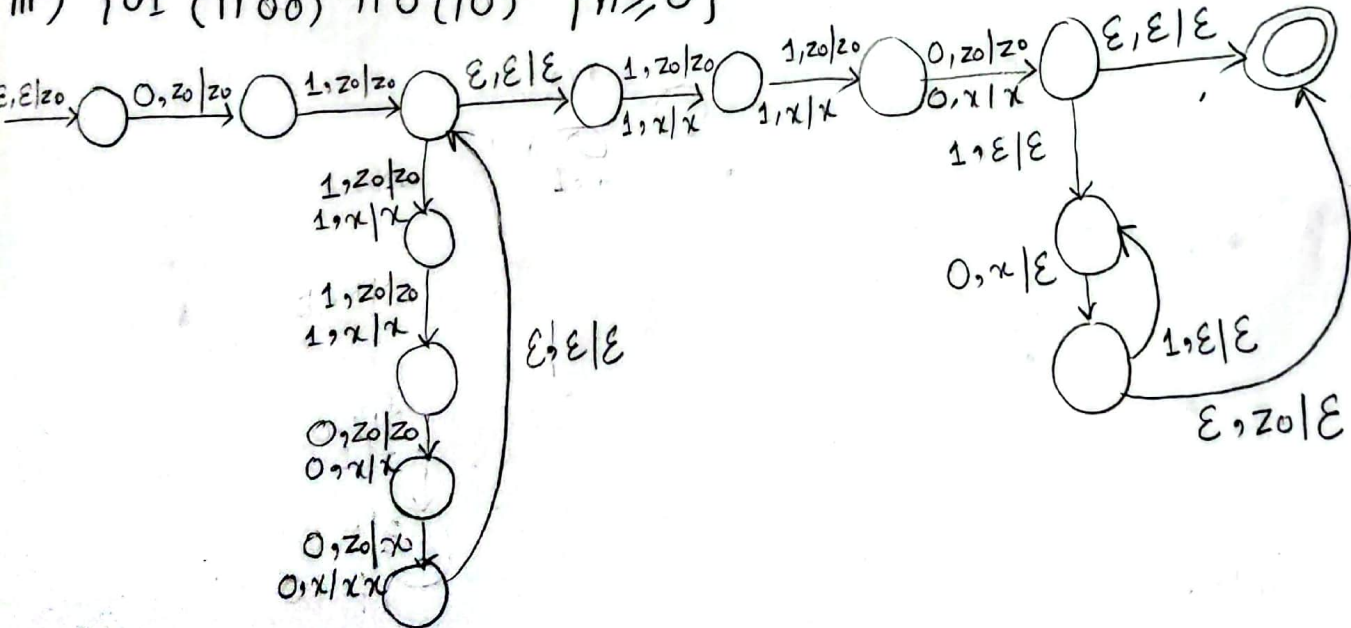


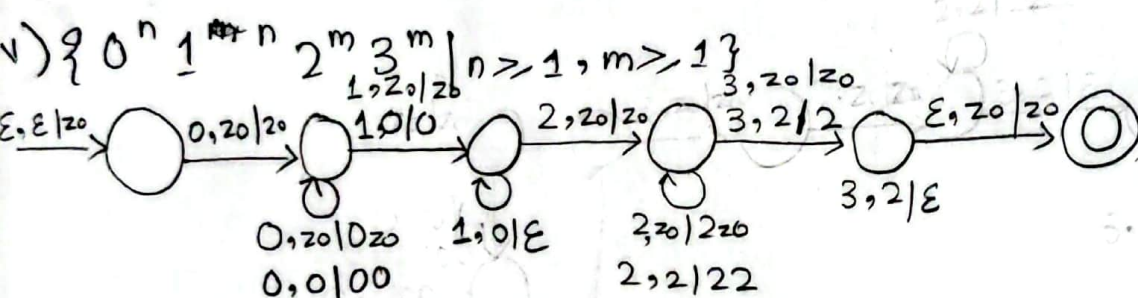
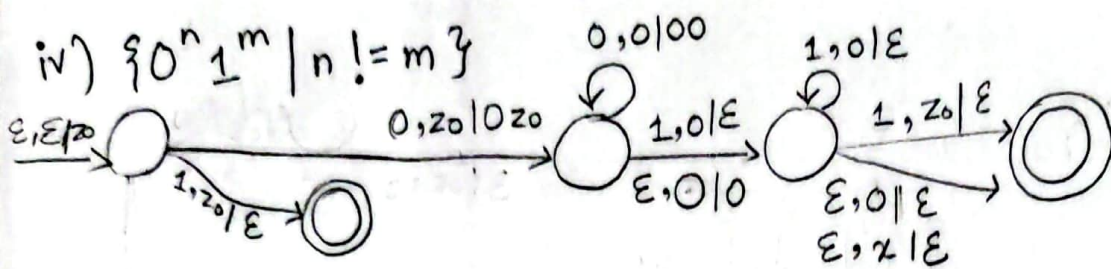
0, 2, 0 0 2, 0	0, 0 8
1, 2, 0 1 2, 0	1, 1 8
0, 0 0 0	
1, 0 1 0	
0, 1 0 1	
1, 1 1 1	

b) odd palindromes.


$$\begin{array}{l} 0, z_0 | z_0 \\ 1, z_0 | 1z_0 \\ 0, 0 | 00 \\ 0, 1 | 01 \\ 1, 0 | 10 \\ 1, 1 | 11 \end{array}$$

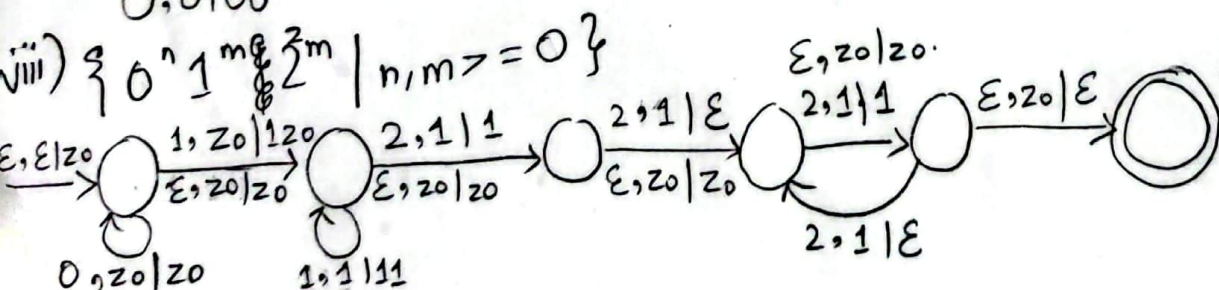
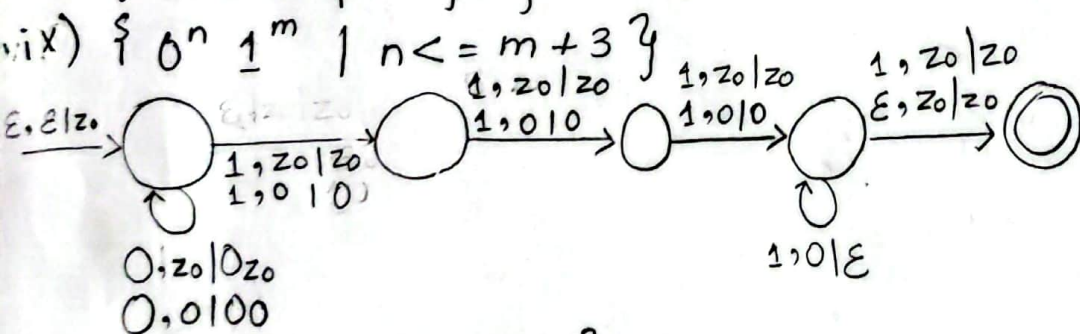
iii) $\{01(1100)^n 110(10)^n \mid n \geq 0\}$





vi) $\{0^i 1^j 2^k \mid k \leq i, k \leq j\}$ unsolvable

vii) $\{0^i 1^i 2^k \mid i = j, j = k\}$ unsolvable



x) : unsolvable.

Q2) a) $M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$

$Q = \{q_0, q_1\}$

$\Sigma = \{ (,), \{, \}, [,] \}$

$\Gamma = \{ z_0,), \{,], \}$

$q_0 = q_0, z_0 = z_0, F = \{q_1\}$

δ : - x represents whatever was in stack.

$\rightarrow \delta(q_0, (, \epsilon) \rightarrow (q_0,)x)$

$\delta(q_0, [, \epsilon) \rightarrow (q_0,]x)$

$\delta(q_0, \{, \epsilon) \rightarrow (q_0, \{x)$

$\delta(q_0,), \epsilon) \rightarrow (q_0, \epsilon x)$

$\delta(q_0, \}, \epsilon) \rightarrow (q_0, \epsilon x)$

$\delta(q_0,], \epsilon) \rightarrow (q_0, \epsilon x)$

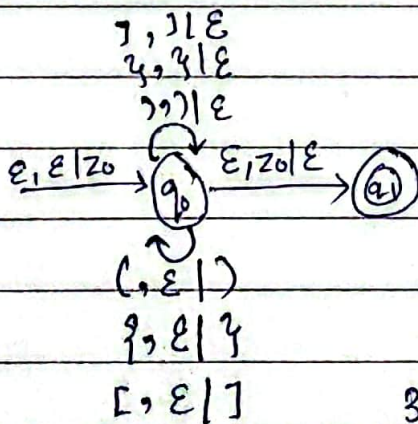
$\delta(q_0, \epsilon, z_0) \rightarrow (q_1, z_0)$

(b) Push: When the PDA reads an opening bracket, it pushes the corresponding closing bracket onto the stack, & pops nothing.

Pop: PDA reads a closing bracket and checks the top of stack for the same bracket. If it matches, then top is popped, else there's no transition and input is rejected.

End Check: If input is fully read & stack has only z_0 , the PDA takes an ϵ transition to accepting state.

(c)



(d) String: $\{ [()] \}$

1. Start:

$\{ [()] \}$, Stack: z_0

2. Read {:

$[()] \}$, Stack: $\{ z_0$

3. Read [: $()] \}$, Stack: $] \{ z_0$

4. Read (: $)] \}$, Stack: $)] \{ z_0$

5. Read): $] \}$, Stack: $] \{ z_0$ 6. Read]: $\}$, Stack: $\{ z_0$

7. Read $\}$: ϵ , Stack: z_0

8. Input ends, top of stack is z_0 , transition to q_1 .

e) $V = \{S\}$,

$\Sigma = \{ (,), \{, \}, [,] \}$

$S = S$

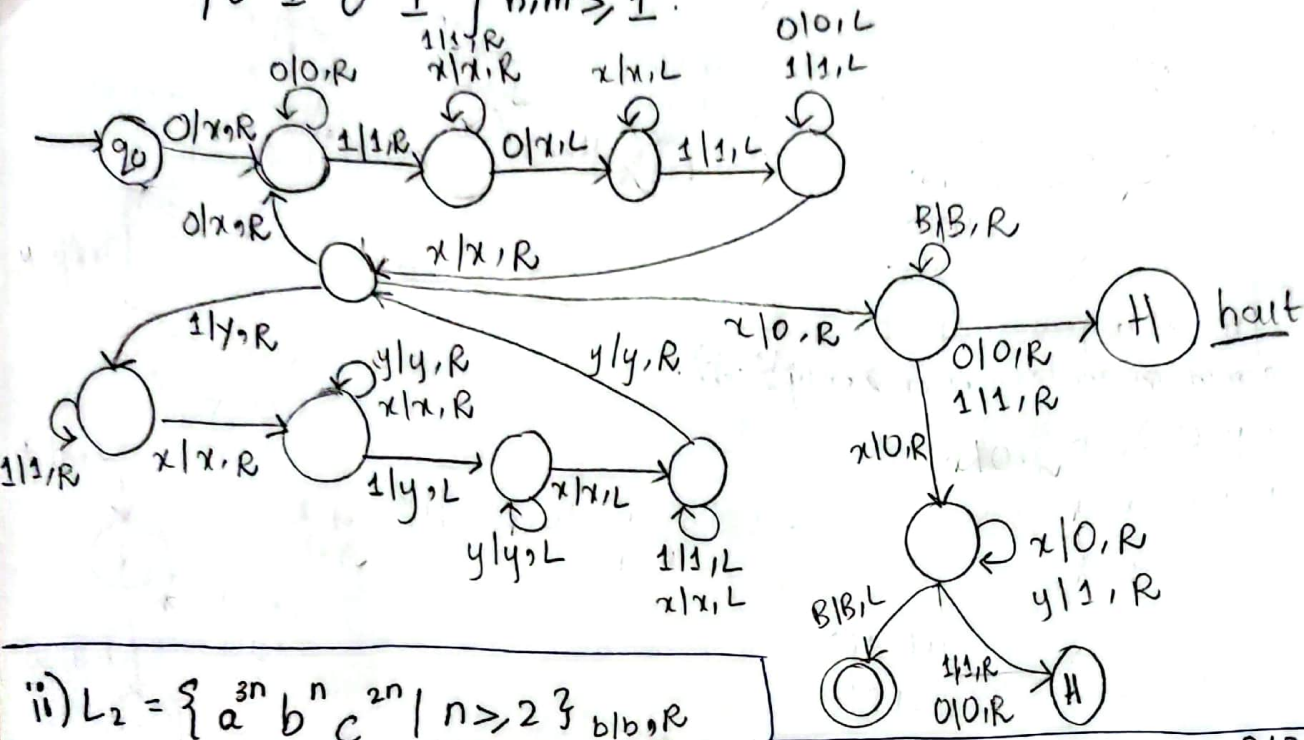
Production Rules:

$$S \rightarrow (S) \mid [S] \mid \{S\} \mid \varepsilon \mid SS$$

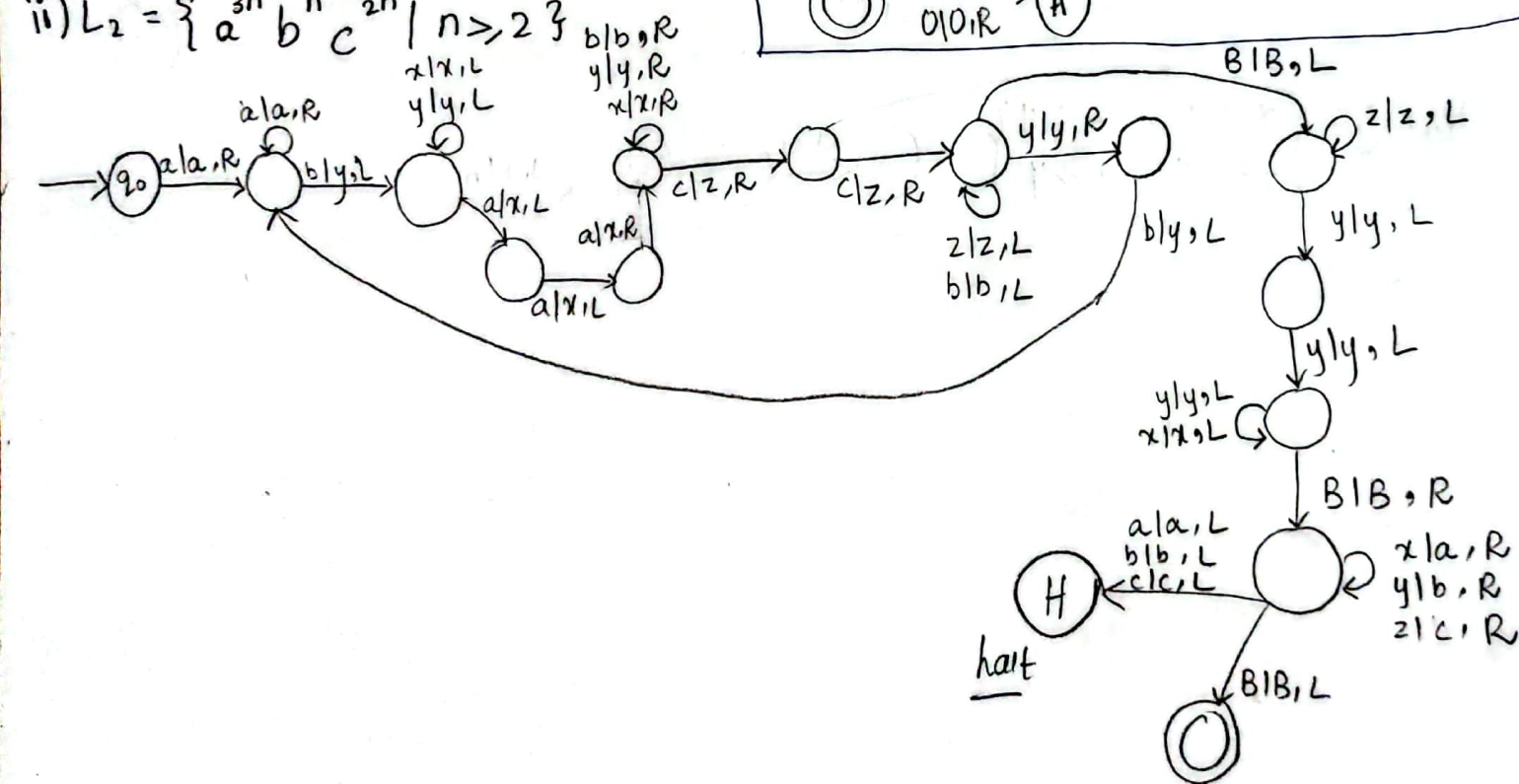
43) TURING MACHINES.

0 0 1 1 1 0 0 1 1 1
x x . y 1 1 x . x y 1 1

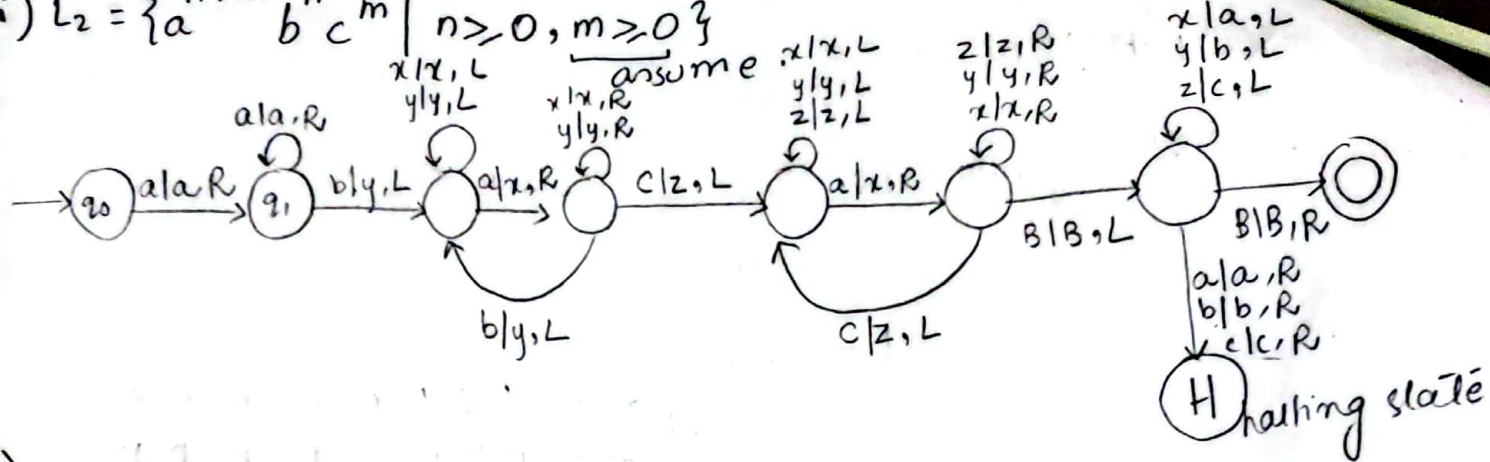
(i) $L = \{0^n 1^m 0^n 1^m \mid n, m \geq 1\}$



(ii) $L_2 = \{a^{3n} b^n c^{2n} \mid n \geq 2\}$

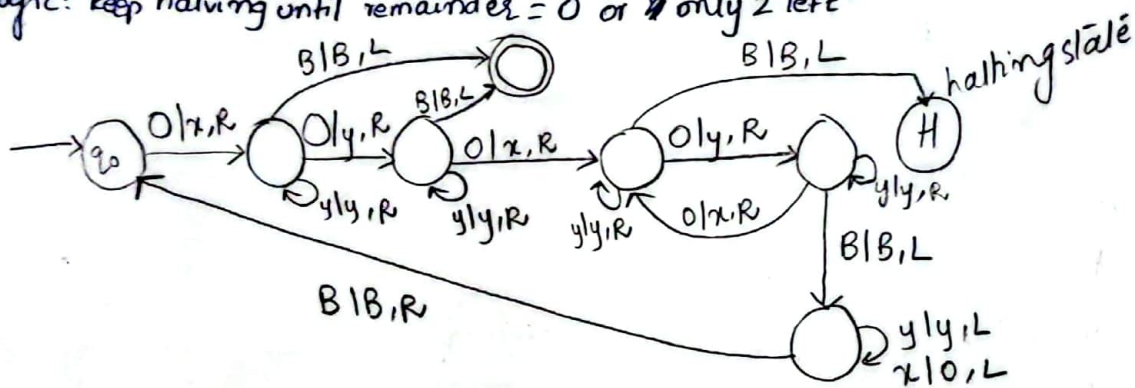


iii) $L_2 = \{a^{n+m} b^n c^m \mid n \geq 0, m \geq 0\}$

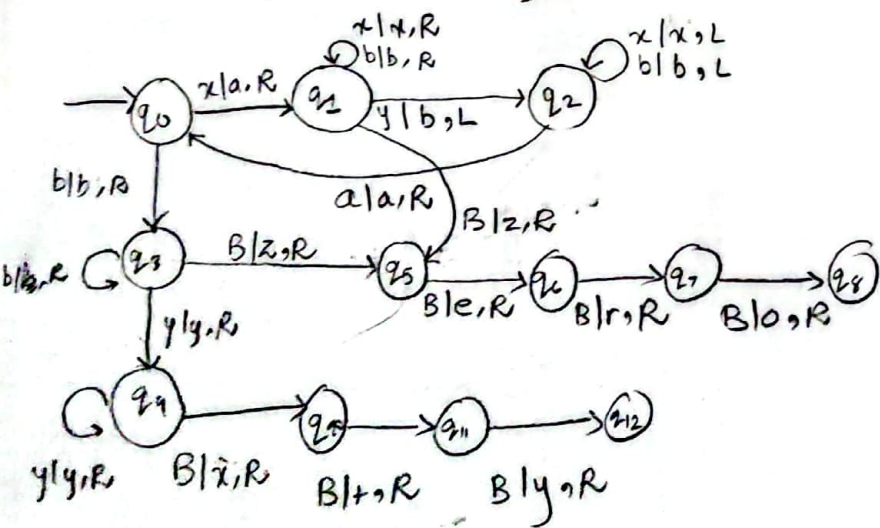


iv) $L_3 = \{0^n \mid n \text{ is a power of } 2\}$.

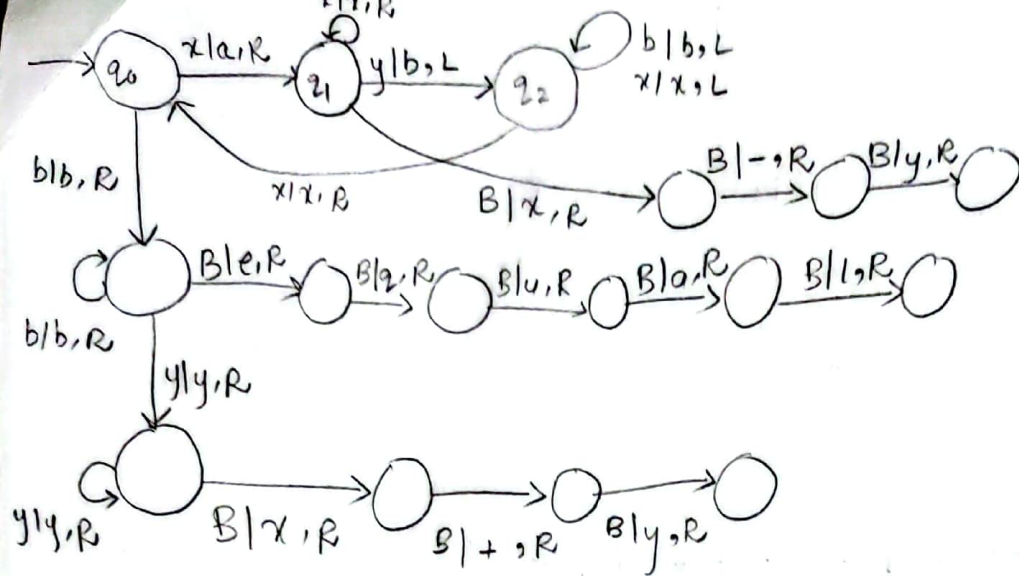
logic: keep halving until remainder = 0 or only 2 left.



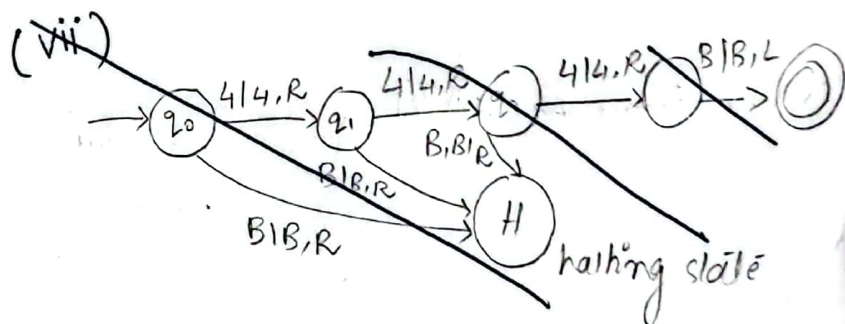
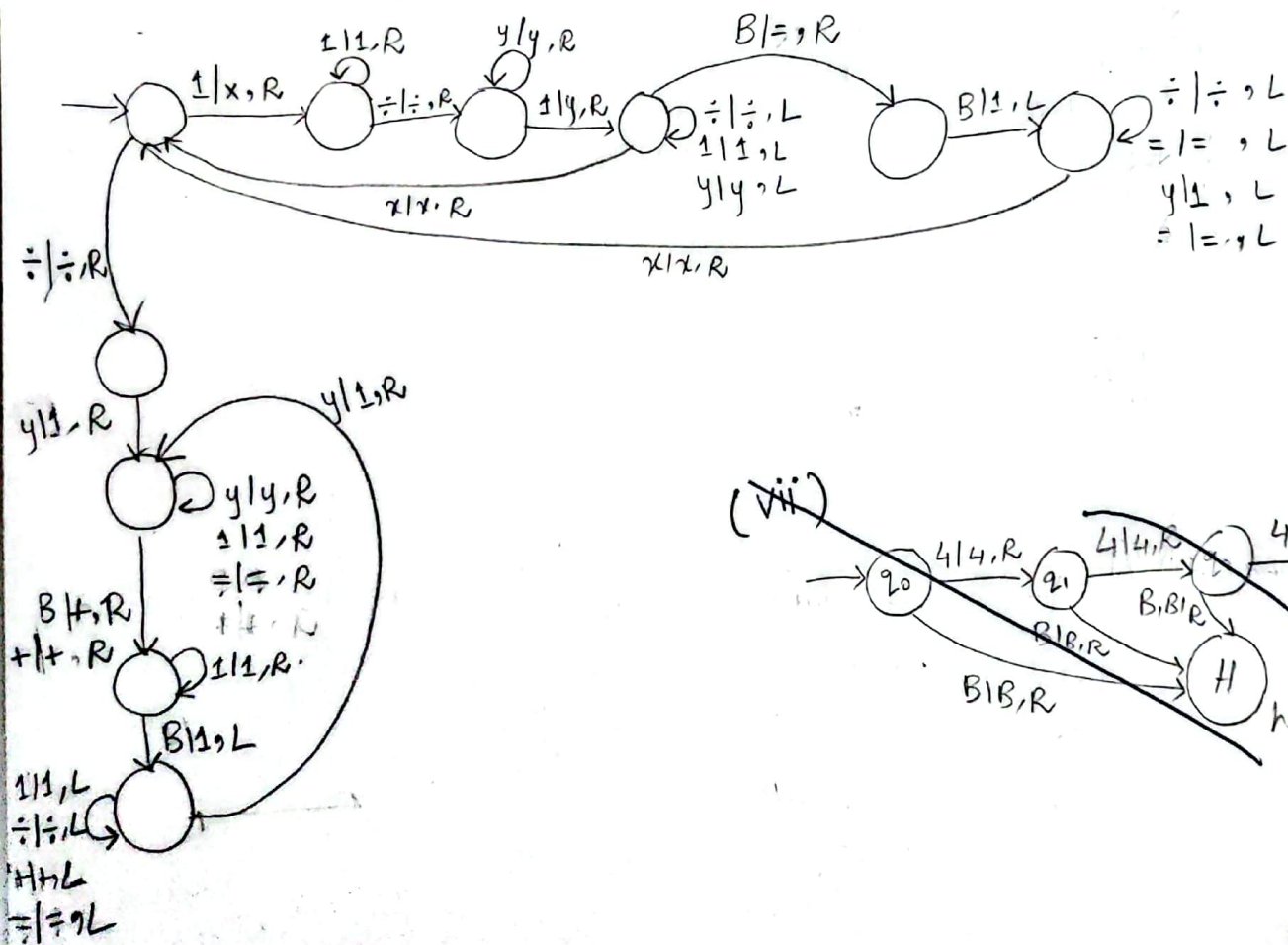
$$v) f(x, y) = \begin{cases} x + y & x < y \\ \text{"zero"} & x \geq y \end{cases}$$

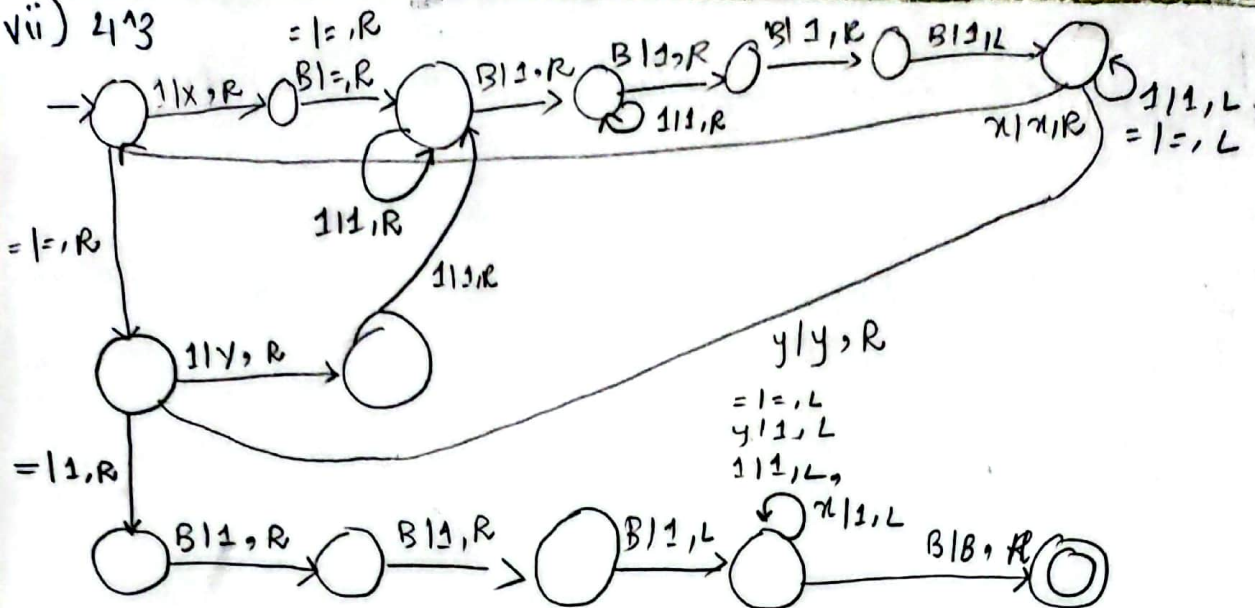


$$d = \begin{cases} x+y \\ x-y \end{cases} \quad \begin{matrix} x < y \\ x > y \end{matrix} \quad \text{else "equal" on tape. (vi)}$$

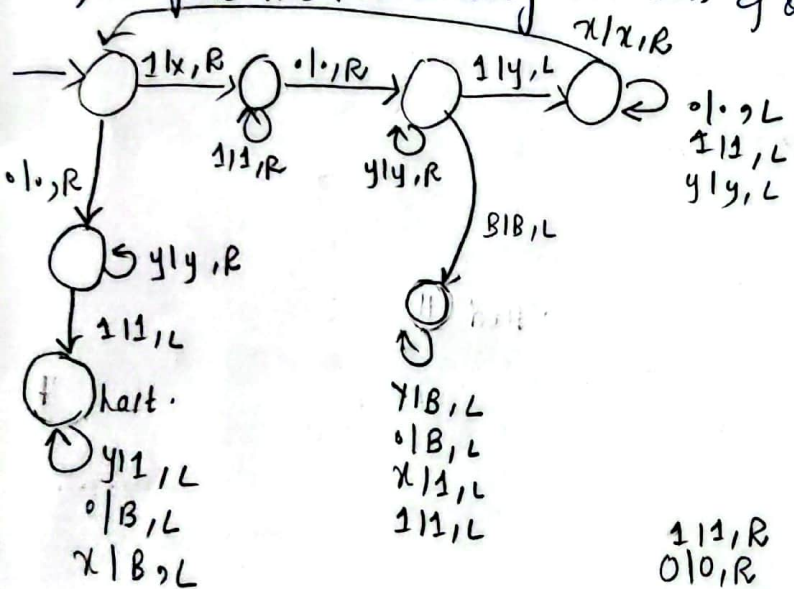


viii) divide two unary numbers, output quotient + remainder.



$$v_{ii}) \quad 2^3$$


ix) Compare the two unary numbers $x/x, y/y$ & output the largest on tape.



x) Palindrome ~~for~~ (odd) TM:

